

Purnima Bhattra

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Frontend Developer & UI/UX Designer

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Lalitpur, Nepal

[Portfolio](#)

EDUCATION

Nepal College of Information Technology, Pokhara University
Bachelor Engineering in Computer Engineering (BECE) – 7th Semester

Lalitpur, Nepal
2023 – Present

TECHNICAL PROJECTS

NestFinder: AI-Powered Room Finding Platform

React, Leaflet, Supabase, Flask, PostgreSQL

UI and frontend lead (two-person team)

2026

- Designed and build tenant, landlord and admin dashboards with responsive UI.
- Integrated Leaflet/OpenStreetMap for map-based room discovery and location visualization.
- Implemented authentication, role-based navigation, room listing and inquiry workflows.
- Contributed to AI recommendation integration, testing, project documentation and presentation.
- Live: [NestFinder](#) Code: [GitHub](#)

Biometric-Based Electronic Voting Machine

Esp32, C++, IOT

Team lead, Presenter and hardware developer (four-person team)

2026

- Build a low-cost voting machine as an alternative to commercial systems priced beyond developing countries' reach.
- Two-step authentication: officer database check, then fingerprint confirmation with single-vote enforcement.
- Stored ballots on detachable memory, so results can be carried independently once polling closes.
- Demonstrated for three days at LOCUS 2026, Nepal's largest tech festival, receiving news-agency coverage.
- Code: [GitHub](#)

Thyroid Disease Prediction: Random Forest vs. Neural Network

Python, scikit-learn, Keras

Machine Learning Developer (course project)

2026

- Compared Random Forest against a tuned neural network for three -class thyroid classification (UCI ANN -thyroid dataset).
- Random Forest reached 99.5% accuracy and 98.34% macro F1, far outperforming the tuned ANN (80.49% / 62.84%).
- Tuned and evaluated Random Forest using stratified splitting and classification metrics.
- Performed preprocessing, hyperparameter tuning and model comparison on imbalanced data.
- Code: [GitHub](#)

NepArt Creations: Handicraft E-Commerce Marketplace

PHP, MySQL

UI/ Frontend Developer (four-person team)

2025

- Designed and developed responsive interfaces for a Nepali handicraft e-commerce platform.
- Code: [GitHub](#)

Additional Projects

Developer

- Basket-Ball Dribble Simulation: continuous event simulation with live gravity, energy loss after each bounce and collision count. Code: [GitHub](#)
- Line following car: detects color line and follows it; build at LOCUS 2026.

TECHNICAL SKILLS

Languages: JavaScript, Python, SQL, Java, C, C++.

Front-End: React, HTML5, CSS3, responsive and cross-browser design.

Backend & Databases: MySQL, Supabase, PostgreSQL.

UI/UX: Dashboard and interface design, responsive layouts, user flows, role-based navigation.

Hardware and IoT: ESP32, Arduino, embedded C++, sensor integration, biometric modules, circuit design.

Tools and Platforms: Git, GitHub, Figma, Vercel, VS Code, Arduino IDE, Linux.

Spoken Languages: Nepali(native), English(proficient), Hindi(proficient).

REFERENCE

Amit K. Shrivastava

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